

BOX 2026 - Homework 2

Joachim Le Fournis

Remarks on the notes :

When we encounter non-usual bases (such as Y or N), we decided to split the read sequences as two sequences, in order to avoid creating non-existent k-mer.

2.

3.

Filename	Number of sequences	Cummulative length of sequences	Species found
file1.fa	1	2221315	Streptococcus pneumoniae
file2.fa	1	29903	Severe acute respiratory syndrome coronavirus 2
file3.fa	1	9181	Human Immunodeficiency Virus (HIV)
file4.fa	7	2553358	Different results depending of the sequence : Streptococcus hyointestinalis, no results on other sequences, ...
file5.fa	1	29706	Severe acute respiratory syndrome coronavirus 2
file6.fa	1	213008	Streptococcus suis strain

4.

Jaccard indexes	file1.fa	file2.fa	file3.fa	file4.fa	file5.fa	file6.fa
file1.fa	1.0	9.17e-07	0.0	0.0033	0.0	2.64e-05
file2.fa	9.17e-07	1.0	0.0	0.0	0.82	0.0
file3.fa	0.0	0.0	1.0	0.0	0.0	0.0
file4.fa	0.0033	0.0	0.0	1.0	0.0	0.088
file5.fa	0.0	0.82	0.0	0.0	1.0	0.0
file6.fa	2.64-05	0.0	0.0	0.088	0.0	1.0

As expected, most of the cells are filled with 0, and 1 on the diagonale (because most of the genomes are different, and every genome is the same as itself). However, we also find that both file2.fa and file5.fa are very similar (82%), as expected because both of them matches Severe acute respiratory syndrome coronavirus 2. Also, file4.fa, file1.fa and file6.fa have similarities (non-null Jaccard indexes), probably because all of them are Streptococcus.

6.

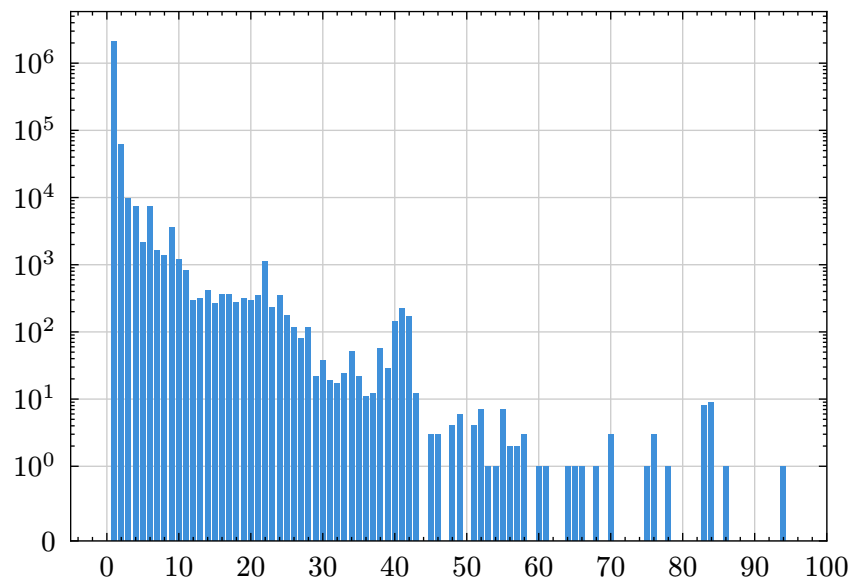


Figure 1: canonical k – mer abundances with $k = 40$ for file4.fa